LaTeX File Template

A guide to its setup, formatting, and custom commands.

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This version of the documentation is accurate as of Version 3.0.0 of the template.

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If there are any issues, contact me using one of the below options:
GitHub or LinkedIn.

Abstract

Thank you for choosing to use LaTeX File Template. If you find any bugs or want some feature not currently included, then my contact information can be found on my GitHub page. For the avoidance of doubt, while my time at Durham University may be mentioned in places in this document, this is an independent project, and as such is in no way associated with the univerisity.

This template was designed to create a structured starting point for academics (especially those studying mathematics or computer science) to create professional, well-formatted documents that follow traditional dissertation guidelines. This repository contains the actual template, documentation files (linked below), and their supporting build files, demonstrating the capabilities of the template. This version of the documentation is accurate for Version 3.0.0 of the template.

Acknowledgements

I would like to thank Dr Jamie Mason for his support and expertise in leading my discrete mathematics tutorials and seminars during my first year at Durham University, without which this project would probably not exist. His continuous feedback throughout has allowed me to hone the project from the messy original versions into a useable tool.

Thanks also to Joseph Eddon for supporting the development of this template through the creation of his own LaTeX template, and for the feedback and suggestions he provided on multiple versions of the template throughout last year.

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Contents

1 Template Setup **1**
- 1.1 Choosing your Compiler . . . 1
  - 1.1.1 LaTeX Workshop . . . 1
- 1.2 The Template Files . . . 1
  - 1.2.1 The Main Files . . . 1
  - 1.2.2 The Template Setup Folder . . . 2
  - 1.2.3 The Chapters Folder . . . 2
  - 1.2.4 Additional Files . . . 2
- 1.3 Packages List . . . 2

2 Formatting Options **3**
- 2.1 Document Formatting . . . 3
  - 2.1.1 The Title Page . . . 3
  - 2.1.2 Changing the Colour Scheme . . . 3
- 2.2 Additional Sections . . . 3
  - 2.2.1 The Prelude . . . 3
  - 2.2.2 The Appendix . . . 3

3 Custom Commands **4**
- 3.1 Text Formatting . . . 4
  - 3.1.1 Custom Date Formats . . . 4
  - 3.1.2 The Colour Scheme . . . 4
- 3.2 General Commands . . . 4
  - 3.2.1 Figures and Tables . . . 4
  - 3.2.2 Academic Environments . . . 4
  - 3.2.3 Environment References . . . 4
  - 3.2.4 Citations . . . 4
- 3.3 Mathematics and Computer Science . . . 5
  - 3.3.1 Additional Mathematics Commands . . . 5
  - 3.3.2 Computer Science Environments . . . 5

4 The Bibliography **6**
- 4.1 Formatting in your Document . . . 6
  - 4.1.1 Citation Format . . . 6
  - 4.1.2 Bibliography Format . . . 6
- 4.2 The Bibliography File . . . 7
  - 4.2.1 Entry Formatting . . . 7
  - 4.2.2 Field Formats . . . 8

5 Appendix **9**
- 5.1 Bibliography . . . 9
- 5.2 List of Tables . . . 9
- 5.3 List of Figures . . . 9

Chapter 1

Template Setup

1.1 Choosing your Compiler

When designing this template, I have exlusively worked in Visual Studio Code, due to its good debugging features, access to the build files, and syntax highlighting. As such, all the formatting and commands of this template have been extensively tested in this compiler, and as it stands, have not been tested elsewhere. This means that, until I have the opportunity to test this elsewhere, I would suggest using it in Visual Studio Code as well.

1.1.1 LaTeX Workshop

To get LaTeX code compiling in Visual Studio Code, I used the extension LaTeX Workshop (Yu et al., 2026). To set this module up, the link in the bibliography is to their project page, where they have outlined installation instructions.

1.2 The Template Files

To reduce the technical understanding required to use this template, I have split the build files across a few folders to hide files that don't need to be edited, or to group those that make logical sense together. Below, I have explained these groups and how each file works so that you can understand the workings of the template if you so wish.

1.2.1 The Main Files

Outside of the folders, there are three files in the root folder of the template, of these, one is .gitignore which is explained in Section 1.2.4 under the heading *Working with Github*. The other two files are file-name.tex and file-name.pdf.

Document Compiler: file-name.tex

This is the *.tex* file that compiles the template into the output PDF file-name.pdf. To edit the name of the output PDF, you simply need to rename this file and change the file name in line one of this file.

After reviewing all the *documentclass* types LaTeX provides, I chose to adopt the *report* type with the default font size being 11pts. This class gives a good balance between the simplicity of some of the other more basic academic-style templates, and the modularity of some of the classes designed for longer documents.

This choice is best illustrated by the *Chapter* divider, which lends itself to the file structure of the template, allowing for long documents to be created with clear separation between parts beyond the default sections and their sub-types.

Template Output: file-name.pdf

This is the output file of the template. All of the content added to the various Chapter files will be compiled and placed in here. To change the appearance or content of this file, you will need to edit one of the other files making up the template as detailed in the rest of this guide.

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1.2.2 The Template Setup Folder

1.2.3 The Chapters Folder

1.2.4 Additional Files

Working with GitHub

I would always advise working with GitHub when writing LaTeX projects as a good method of version control. As such, I have placed a .gitignore file into the template folder which will ignore any/all build files generated by the compiler (tested for LaTeX Workshop).

Inserting Figures

If you wish to insert figures/images into your document, then they should be placed in the Figures folder. This will allow you to reference them directly when inserting them into your project (i.e., use image_name rather than Figures/image_name). For instructions on inserting images into your project, the command is documented as INSERT COMMAND.

Creating a Bibliography

If you need your project to include a bibliography, then the functionality for this are already in place within the document, but you will need to edit some of the source code relevant to it. Instructions for this can be found in Chapter 4.

1.3 Packages List

To create this template, I had to select a variety of packages to be used throughout to provide additional functionality for formatting and custom commands. Below is a list of the packages installed, what they do, any additional documentation required, and credit to the original creators. Add the list of packages.

Chapter 2

Formatting Options

2.1 Document Formatting

2.1.1 The Title Page

2.1.2 Changing the Colour Scheme

2.2 Additional Sections

2.2.1 The Prelude

2.2.2 The Appendix

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Chapter 3

Custom Commands

3.1 Text Formatting

3.1.1 Custom Date Formats

3.1.2 The Colour Scheme

3.2 General Commands

3.2.1 Figures and Tables

3.2.2 Academic Environments

3.2.3 Environment References

3.2.4 Citations

For this template, I have chosen to setup my citation commands inline with Harvard-style citation guidelines. As such, I have provided are two different formats for inserting a citation, direct and indirect. Their use-cases and implementations are described below.

Command 3.2.4.1: A direct citation is used when referring to a bibliography entry directly, for example when saying "according to *bibliography_entry...*". Given the Harvard-style guidelines, this follows the general format:

author, author, ..., and author (year_published)

To call this function, we use the syntax given by Algorithm 1.

Algorithm 1 Direct citation format.

```
\directcitation{bibliography_entry_key}
```

The parameters for this function are shown below in Table 1.

Parameter	Description	Example
bibliography_entry_key	The key given to the entry when it was placed in references.bib.	biblatex_manual

Table 1: Direct citation parameters.

Command 3.2.4.2: An indirect citation is used when referring to a bibliography entry as support for a statement, for example when saying "according to the paper (*bibliography_entry*)...". Given the Harvard-style guidelines, this follows the general format:

(author, author, ..., and author, year_published)

To call this function, we use the syntax given by Algorithm 2.

Algorithm 2 Indirect citation format.

```
\indirectcitation{bibliography_entry_key}
```


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The parameters for this function are shown below in Table 2.

Parameter	Description	Example
bibliography_entry_key	The key given to the entry when it was placed in references.bib.	biblatex_manual

Table 2: Indirect citation parameters.

3.3 Mathematics and Computer Science

3.3.1 Additional Mathematics Commands

3.3.2 Computer Science Environments

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Chapter 4

The Bibliography

The template is set up with an automatically populating bibliography using BibLaTeX. This allows you to maintain and organise your citations all in one place with consistent formatting being created by the library and my formatting settings. At the moment, the template does not contain any citations, and to prevent errors, I have had to comment out some lines. These are: line 10 on file-name.tex (shown below as Algorithm 3),

Algorithm 3 Commented bibliography lines (file-name.tex).

```
\addbibresource{./Template-Setup/references.bib} % Uncomment to add a bibliography.
```

and lines 5-9 on Chapters/appendix.tex (shown below as Algorithm 4).

Algorithm 4 Commented bibliography lines (Chapters/appendix.tex).

```
\begingroup
\section{Bibliography}
\defbibheading{bibliography}[]{}
\printbibliography
\endgroup
```

However, as soon as you want to add a citation, simply uncomment the lines, and then insert it into the bibliography as described in Section 4.2.

4.1 Formatting in your Document

4.1.1 Citation Format

To ensure that the citations are standardised throughout the document, I have chosen to default to Harvard style citations for the project (or as close as is possible). As such, when citing one of the bibliography entries there are two in-text options, depending on how you are referring to it.

- If you are referencing something directly, you should use the format *author, author (year_published)* (e.g., according to author (year_published)). This is implemented using the command described in Command 3.2.4.1.
- Conversely, if you are referencing something indirectly, you should use the format *(author, year_published)* (e.g., according to research (author, year_published)). This is implemented using the command described in Command 3.2.4.2.

4.1.2 Bibliography Format

When designing this version of the template, I chose to make the bibliography as customisable as I could, whilst maintaining the simplicity of the system. As such, the bibliography has been set up to be completely unsorted. This allows the user to choose how to sort it, rather than forcing them into a set version. The drawback of this system is that the user does then have to sort the entries themselves but, on balance, I believe that this is a good tradeoff.

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4.2 The Bibliography File

Entries into the bibliography are stored in Template-Format/references.bib and are then automatically pulled through into the bibliography (in the appendix). To add a new entry, follow the guide below about the entry formats and the general rules around entries. Please note that this is not a definitive list and can be adapted using the more comprehensive guide at Kime, Wemheuer and Lehman (2026).

4.2.1 Entry Formatting

BibLaTeX provides a variety of entry types, for simplicity's sake, I have listed them below, alongside a description of their use, and their required fields for Harvard references as seen in the tables below. This was produced in combination between OpenAI (2026) and Kime, Wemheuer and Lehman (2026).

Source Type	Entry Type	Required Fields	Optional Fields
Book	@book	title, author, date.	editor, edition, chapter, pages, publisher, location, isbn, doi, url, urldate.
Dictionary/Encyclopedia Entry	@inreference	author, booktitle, title, date.	editor, edition, chapter, pages, publisher, location, isbn, doi, url, urldate.
Film/Movie	@movie	title, author, date.	publisher, location, url, urldate.
Interview	@misc	author, date, howpublished.	title, organization, url, urldate.
Letter	@letter	author, date, howpublished.	title, organization, url, urldate.
Manual/Documentation	@manual	title, author, date.	version, url, urldate.
News Article	@article or @online	title, author, journaltitle, date.	organization, pages, url, urldate.
Speech	@unpublished	author, date, eventtitle.	title, venue, organization, url, urldate.

Table 3: biblatex .bib file entry formats (traditional media).

Source Type	Entry Type	Required Fields	Optional Fields
AI Chat	@online	author, title, date, note (with model), url, urldate, organization.	n/a.
Blog Post	@online	author, title, date, url, urldate.	organization.
Podcast/Audio Recording	@audio	author, title, date, url, urldate.	publisher, series, number.
Social Media Post	@online	author, date, organization, url, urldate.	n/a.
Software	@software	author, title, version, date.	organization, url, urldate.
Website	@online	author, title, date, url, urldate.	n/a.

Table 4: biblatex .bib file entry formats (online).

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Source Type	Entry Type	Required Fields	Optional Fields
Legal Case	@jurisdiction	author, title, date, institution.	url, urldate.
Legislation	@legislation	title, date.	section, institution, url, urldate.
Patent	@patent	author, title, date, number.	location, organization, url, urldate.

Table 5: biblatex .bib file entry formats (legal/political).

Source Type	Entry Type	Required Fields	Optional Fields
Article (Journal)	@article	author, title, date.	journaltitle, volume, number, pages, doi, url, urldate.
Conference Paper	@inproceedings	author, title, date, eventtitle.	venue, pages, publisher, pages, doi, url, urldate.
Dataset	@dataset	author, title, date.	publisher, version, doi, url, urldate.
Document (General)	@misc	author, title, date.	howpublished, publisher, location, number, url, urldate.
Presentation/Conference Talk	@unpublished	author, title, date, eventtitle.	venue, organization, url, urldate.
Report	@report	author, title, date.	organization, number, type, publisher, location, url, urldate.
Thesis/Dissertation	@thesis	author, title, institution, date.	doi, url, urldate.

Table 6: biblatex .bib file entry formats (academic).

4.2.2 Field Formats

OpenAI (2026) and Kime, Wemheuer and Lehman (2026) also defines the data types and formats of each of the fields. Below, I have listed a summary of the formats and their rules. An example of a complete entry can be seen in Algorithm 5.

- Citations should be identified with unique keys – ideally *title_description*.
- All fields are surrounded in curly brackets.
- Where the author is an organisation, you may need to add an additional set of curly brackets to ensure it is kept together as one entry.
- Multiple authors should be separated with 'and's.
- Dates are formatted as YYYY-MM-DD.

Algorithm 5 biblatex .bib entry format.

```
@entry_type{bibliography_entry_key,
    author = {author_one and author_two...},
    title = {piece_title},
    date = {date_published},
    version = {piece_version},
    url = {piece_url},
    urldate = {date_accessed}
}
```


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Chapter 5

Appendix

5.1 Bibliography

Yu, James et al. (Sept. 2026). *LaTeX Workshop*. Version 10.19.0. URL: https://github.com/James-Yu/LaTeX-Workshop/wiki (visited on 12/10/2026).

Kime, Philip, Moritz Wemheuer and Philipp Lehman (2026). *The biblatex Package*. Version 3.22a. URL: https://tex.org.uk/macros/latex/contrib/biblatex/doc/biblatex.pdf (visited on 10/10/2026).

OpenAI (2026). *BibLaTeX Reference Guide for Harvard-style Citations*. Model is GPT-6 Luna. OpenAI. URL: https://chatgpt.com/s/t_6aca1d00deb48191a4a0d9e4498cd5ea (visited on 10/10/2026).

5.2 List of Tables

1 Direct citation parameters. 4
2 Indirect citation parameters. 5

3 biblatex .bib file entry formats (traditional media). 7
4 biblatex .bib file entry formats (online). 7
5 biblatex .bib file entry formats (legal/political). 8
6 biblatex .bib file entry formats (academic). 8

5.3 List of Figures

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